



7.1 Student Guide: Introduction to Windows and CMD

Class Overview

Today's class begins our module on Windows administration and hardening. You will be introduced to the Windows operating system and command line by performing basic system administration tasks today.

Class Objectives

By the end of class, you will be able to:

- Leverage the Windows Command Prompt (CMD) to execute various sysadmin responsibilities.
- Use `wmic` and Task Manager to manage applications and services.
- Create, manage, and view user information using the command-line tool `net`.
- Manage password policies using `gpedit`.
- Optionally, schedule tasks using Task Scheduler.

Lab Environment

Lab Details

Open the Hyper-V Manager in the Windows RDP Host machine to access the nested virtual machines:

- Windows 10 machine
 - Username: sysadmin
 - Password: cybersecurity
- Windows Server machine
 - Username: sysadmin
 - Password: p4ssw0rd*

The instructor lectures, demonstrations, student activities, and reviews will all be completed using the Windows RDP Host Machine.

We won't need the nested Hyper-V virtual machines until the third day of this module.

Module Day 1 Content Tracker

1. [Introduction to Windows](#)
2. [Launching Your Windows Lab](#)
3. [Intro to Task Manager](#)
4. [Windows File Structure](#)
5. [Windows OS Folders](#)
6. [Windows Users](#)
7. [Introduction to Command Prompt CMD](#)
8. [Student Do: Intro to Task Manager and CMD Activity](#)
9. [Review: Intro to Task Manager and CMD Activity](#)
10. [Break](#)
11. [Users, Groups, and Password Policies](#)
12. [Users, Groups, Password Policies](#)
13. [Review: Users, Groups, Password Policies](#)
14. [Creating Users and Setting Passwords](#)
15. [Create Users and Set Password Policies](#)
16. [Review: Create Users and Setting Password Policy](#)
17. [Optional: Task Scheduling](#)

Slideshow

The slides for today can be viewed on Google Drive here: [7.1 Slides](#)

- Note: Editing access is not available for this document. If you or your students wish to modify the slides, please create a copy by navigating to File > "Make a copy...".

01. Introduction to Windows

Today's lesson will cover fundamental system administration of Windows.

- Do you use Windows machines at home? At work?
 - Note that while many IT professionals prefer Mac OS and Linux for their operating systems, Windows is still the market leader for desktop operating systems.

The widespread use of Windows for personal computers and workstations makes them the most common target for today's attackers.

- Much of the malware that exists today specifically targets vulnerabilities in unpatched and unsecured Windows personal computers and servers.
- Therefore, as a cybersecurity professional, you will likely interact with Windows — whether it's the operating system you use daily, or as part of your company or client's infrastructure.

Understanding Windows is essential for the following IT and cybersecurity specializations:

- **SOC Analysis:** As an SOC (Security Operations Center) analyst, your familiarity with Windows will be tested almost every day. As part of your daily work, you may observe anywhere from one to several thousand security-related incidents and alerts dealing with Windows endpoints.
- **System Administration:** The large majority of system administrators work with one or more Microsoft products and/or services, including Windows PCs, Windows Servers, Office 365, and Exchange.
- **Penetration Testing:** Due to Windows workstations and servers being widely used in organizations and businesses, it is vital that penetration testers know how to exploit Windows and Microsoft-related products.
- **Endpoint Forensics:** Because the most commonly supported endpoint device in an organization is likely to be Windows, it is critical that forensics investigators have a detailed understanding of how Windows works.

Today's activities and demonstrations will cover common system administration tasks using command-line and GUI tools to troubleshoot a problematic Windows PC. During this process, we will create a report with our findings.

We will complete the following tasks throughout the day:

- Audit processes with Task Manager.
- Use the command line to gather information and create files.
- Enforce password policies.
- Manage users.
- Automate tasks.

Today's Windows activities will move quickly. You will learn the "Windows way" of performing basic system administration tasks that you've already learned to do on Linux.

This gives you a substantial learning advantage. You already know the concepts; you are just learning a new operating system and syntax.

02. Launching Your Windows Lab

Use RDP to log into your Azure lab if you have not yet.

- Credentials for the Azure labs Windows RDP Host machine:
 - Username: `azadmin`
 - Password: `p@ssw0rdp@ssw0rd`

03. Introduction to Task Manager

Did you notice the various processes that started up when you logged into the Windows RDP host machine?

This Windows RDP host machine has some applications installed on it (which will be dealt with in a few moments). To get started, we will need to close the startup application windows.

This is what a Windows workstation can look like if it's not maintained by an organization's system administrators.

- For any application or installer window that launches after booting the machine:
 - Select the application window in the bottom toolbar.
 - Click the top-right **X** in each window to close it.
 - Do not log into any of these applications.
- The following is a list of the applications you should expect to see when booting up the lab:
 - The **Skype** launcher
 - The **Spotify** browser signup page
 - The **Spotify** login screen

We'll look at uninstalling these applications with a script on our second day of the module.

Remember that during the Linux system administration lessons, you learned how to enumerate and inspect running processes using **top**.

Introducing Task Manager

Today's first demonstration focuses on cleaning up these unnecessary processes with a Windows process management tool: **Task Manager**.

- Windows, like Linux, has the concept of errant/runaway processes.
- We need a way to inspect and troubleshoot process resource utilization in Windows.
- In Windows, we can use the Task Manager GUI to inspect and manage processes, instead of **top**'s CLI.

- Because of this, using Task Manager to manage processes is easier than using **top**.
- Task Manager also allows us to manage other important system components, such as startup processes.

Processes in Windows are much like the processes and PIDs that you worked with in the Linux modules.

Some use cases for Task Manager include:

- Some programs, when left running while not in use, can take up excessive resources or even allow for unwanted remote connections. Some examples are:
 - Google Chrome, which is well-known for its high memory usage.
 - Teamviewer, the remote desktop application, has had critical issues that have left systems extremely vulnerable and accessible from public connections.
- Some processes can cause memory leaks that result in system instability and abrupt system crashes. These crashes are commonly associated with the "blue screen of death."
 - The following URL can be used to learn more about Task Manager and its use:
 - [How to Geek: Windows Task Manager - The Complete Guide](#)

Security professionals can also use Task Manager to audit processes to identify incorrect or malicious processes:

- For example, many cryptojacking processes, or cryptominers, use excessive CPU and/or network resources even while a system is idle.

Task Manager Overview

For this upcoming demo, we'll use Task Manager to identify resource bottlenecks and startup processes that are potential issues within our Windows RDP host machine.

First, we'll go through how to audit processes based on resource utilization.

- Right-click on the taskbar at the bottom of the Windows desktop and choose **Task Manager**. Expand this window as needed.
- With Task Manager launched, make sure you're on the Processes tab.

All running processes on the computer can be seen in this Processes tab.

- Each column header shows the resource utilization of each process: CPU, Memory, Disk, and Network. These can be sorted in ascending or descending order.
- Click the **Memory** resource column header to test the sorting function.

Click the **Performance** tab.

- This tab shows resource utilization visualized in charts. This is useful information for system administrators when there are extended periods of high resource usage.

One of the key functions of Task Manager is ending unwanted processes.

For example, suppose that we notice spikes in CPU usage. We would need to identify the errant process creating these spikes and end it.

In our case, we have a process that may be considered a security risk.

Task Manager Demo: Ending Processes

This Windows RDP host machine has a process we have identified to be a potential security risk: **TeamViewer**.

TeamViewer is a remote desktop application that is used similarly to RDP to connect to remote computers.

Look through the following URL to learn more:

- [threatpost: TeamViewer Flaw in Windows App Allows Password-Cracking](#)

This process, TeamViewer, has had recent and past vulnerabilities that were deemed critical.

- With Task Manager still open, do the following on the Processes tab:
 - Click the **Process** column header to show all the running processes.
 - Click the **Name** column header to sort all processes alphabetically.
 - Finally, click on **TeamViewer**.

Due to the security risk of having TeamViewer running, we'll want to end its process.

- With the TeamViewer process highlighted, right click and **End task**.

Disabling Startup Processes

Now that we've seen how to stop a process, we'll take a look at how to disable processes from automatically starting up when our machine boots up.

- With Task Manager still open, go to the **Startup** tab:
 - Each listing under the Startup tab is a process that boots up when the machine does.
- Select the **Spotify** startup process.

This is not a process we want starting up anymore.

- Right click and **Disable**. This will prevent the process from starting up with the Windows machine.

Next time this machine is booted, this process will no longer start up with the machine.

While troubleshooting with Task Manager seems like a simple process, being able to manage running and startup processes is an important skill to have across all platforms. It is also a vital first step to solving many security-related problems, such as incident response, digital forensics, and threat hunting.

Managing startup applications is important for system and security administrators for multiple reasons:

- Startup applications can slow boot time due to their execution priority.
- These applications may use excessive resources while in the background, causing random system slowdowns.
- Applications might use the network in the background. They can, for example, initiate their own automatic updates that hog network bandwidth, and become a security risk by opening ports to listen to.
- Startup applications may require special permissions for their functionality. These can pose security risks if, for example, they are compromised through malware. Malware can then potentially run these rogue processes as administrators.

04. Windows File Structure

Before explaining how to do common system administration tasks, it's important to understand how Windows stores the core operating system files. Similar to Linux, Windows uses a folder system on a drive with subdirectories. In Windows, each storage device is represented by a drive letter (e.g., C:, D:, E:). The primary drive, typically the one where the operating system is installed, is often labeled as the C: drive. Additional drives, such as secondary hard drives or external storage devices, are assigned subsequent letters.

Directories (Folders): Directories, also known as folders, are containers used to organize files and other directories. Directories can contain files, subdirectories, or both. The root directory, represented by "", is the top-level directory on each drive. This is different from Linux, where the root directory is represented with "/" (forward slash vs. back slash).

- Use the following default Windows directory structure as an example to help guide through the next topics:

Unset

C:\

└─ PerfLogs\

└─ Program Files\

└─ Program Files (x86)\

└─ ProgramData\ [hidden folder]

└─ Users\

| └─ [username]\

| └─ Desktop\

| └─ Documents\

└─ Windows\

 └─ System32\

 └─ Config\

```
└─ Drivers\  
    └─ etc\  
        └─ hosts  
        └─ networks  
└─ Spool  
└─ Temp
```

Path: A file's path specifies its location within the file system hierarchy. It consists of the drive letter, followed by a series of directory names separated by backslashes (\), and finally the filename. For example, "C:\Users\John\Documents\example.txt" is the path to a file named "example.txt" located in the "Documents" folder within the "John" user's directory on the C: drive.

Special Folders: Windows includes several predefined folders, often referred to as special folders or shell folders, which serve specific purposes. These include folders such as "Documents," "Pictures," "Music," "Downloads," "Desktop," and others. These special folders provide convenient locations for users to store and access commonly used files and data. These folders are created by default whenever a new user profile is created on the machine.

File Attributes: Files in Windows have various attributes that specify properties such as whether the file is read-only, hidden, system, or archived. These attributes control how the file is displayed and accessed by the operating system and applications.

File Extensions: Windows uses file extensions to identify the type of files and associate them with specific applications. For example, ".txt" indicates a text file, ".docx" is a Microsoft Word document, and ".jpg" is an image file in JPEG format. This differs from Linux where file extensions aren't as heavily relied on.

File System: Windows supports different file systems, with NTFS (New Technology File System) being the primary file system used in modern versions of Windows. NTFS provides features such as file and folder permissions, encryption, compression, and journaling for improved reliability.

05. Windows OS Folders

When Windows is installed, it will automatically create several directories and files that are essential to the operating system. It's important as security practitioners to know what these folders and files do, as they can be the target of an adversary or malware.

System32 Folder:

The System32 folder is a critical directory in the Windows operating system. It is located in the `C:\Windows\System32` path on most installations. It contains valuable files that are essential to the OS running, such as:

1. **System Files:** This folder contains essential system files required for the Windows operating system to function properly. These files include dynamic link libraries (DLLs), executables (EXE files), device drivers, and other components necessary for system operation.
2. **System Utilities:** Many important system utilities and tools are located in the System32 folder. These utilities include programs such as `cmd.exe` (Command Prompt), `taskmgr.exe` (Task Manager), `regedit.exe` (Registry Editor), and many others.
3. **32-bit Applications:** Despite its name, the System32 folder on 64-bit versions of Windows contains 64-bit system files, while a separate SysWOW64 folder stores 32-bit system files to maintain compatibility with older applications.

Program Files Folder:

The Program Files folder is another critical directory in the Windows file system. It's located at `C:\Program Files` (or `C:\Program Files (x86)` for 32-bit programs on a 64-bit system).

1. **Application Installations:** When you install new software on your Windows system, it typically gets installed into the Program Files directory by default. Each installed application will have its own subfolder within Program Files, where its files and resources are stored.
2. **Program Data:** In addition to program executables and libraries, the Program Files folder may also contain data files, configuration files, documentation, and other resources associated with installed applications.
3. **User Account Control (UAC):** The Program Files directory is protected by User Account Control (UAC) in newer versions of Windows. This means that administrative privileges are required to modify or delete files within this directory, helping to prevent unauthorized changes to installed programs.

Hidden Folders:

Hidden folders are directories in the file system that have been marked as "hidden," meaning they are not visible by default when browsing files in Windows Explorer or using command-line tools.

1. **Visibility:** By default, Windows hides certain system folders and files to prevent users from accidentally modifying or deleting important system files. These hidden folders include AppData, ProgramData, System Volume Information, and others.
2. **Configuration Files:** Some applications and system processes store configuration files and data in hidden folders to protect them from accidental modification or deletion. For example, the AppData folder within each user's profile directory contains application settings, temporary files, and other data specific to that user.
3. **Viewing Hidden Folders:** While hidden folders are not visible by default, you can choose to show them in Windows Explorer by enabling the "Show hidden files, folders, and drives" option in the Folder Options or View settings. You can also view hidden folders using command-line tools such as `dir /ah` in Command Prompt or PowerShell.

For an adversary or a penetration tester, these hidden files are sometimes sought after as they contain configuration information, which could even include credentials.

06. Windows Users

Windows, like Linux, allows for different user accounts on a machine. These users can have various privileges assigned and can also be targeted by attackers. From the Windows GUI, users are represented as names, but from a functional level, they're represented as security identifiers (SID).

Security Identifier (SID)

A Security Identifier (SID) is a unique alphanumeric string assigned to each security principal (user, group, or computer) in Windows. SIDs are used internally by Windows to identify users and groups and determine their access rights to system resources. They are represented as a variable-length string of alphanumeric characters, typically in the form of a human-readable string, such as "S-1-5-21-3623811015-3361044348-30300820-1013".

- Each SID comprises several components:
 - **Revision Number:** Identifies the revision level of the SID structure. This is typically "S-1".

- **Identifier Authority:** Indicates the authority that issued the SID. For Windows SIDs, this is always "5" for the Security Identifier Authority (SID Authority).
- **Subauthorities:** A variable number of subauthorities that uniquely identify the security principal within the domain or local system. These subauthorities are typically numeric values.

SIDs are designed to be globally unique across all Windows systems. Each SID is unique not only within its own system but also across all systems in a networked environment. Windows generates SIDs using a combination of machine-specific information, such as the unique identifier of the machine, and the SID Authority.

- SIDs are used extensively in access control lists (ACLs) to define permissions on objects such as files, folders, registry keys, and network shares. Each ACE (Access Control Entry) in an ACL contains one or more SIDs representing users or groups and their associated permissions.
- Authentication and Authorization: During the authentication process, Windows uses SIDs to uniquely identify user accounts and groups. Once authenticated, the system references the user's SID to determine their authorization level and access rights.
- Group Membership: SIDs are used to identify members of security groups. When a user is added to a group, their SID is added to the group's membership list. This simplifies permission management by allowing administrators to assign permissions to groups rather than individual users.

Well-Known SIDs:

- Everyone: S-1-1-0, represents all users and groups, including anonymous users.
- Authenticated Users: S-1-5-11, represents all users who have authenticated to the local or a trusted domain.
- Local System: S-1-5-18, represents the built-in SYSTEM account with full system privileges.
- Administrators: S-1-5-32-544, represents the built-in Administrators group.

In essence, SIDs are the cornerstone of Windows security, providing a unique identifier for every security principal in the system and facilitating access control, authentication, and authorization mechanisms. Understanding SIDs is essential for effectively managing security permissions and ensuring the integrity of Windows environments.

SYSTEM

Similar to Linux's 'root' user, Windows has an account that has complete read/write permissions over all files & directories: the 'SYSTEM' account.

- The SYSTEM account is used by the operating system itself and various services that run under the context of the system.
- It has the highest level of privileges on the system, often referred to as "SYSTEM-level privileges."
- This account is essential for the functioning of core Windows services and processes, and it's automatically created during the installation of Windows.
- It's often used to run critical system services, drivers, and processes that require elevated privileges to perform tasks such as managing hardware, interacting with the kernel, and executing critical system operations.

Because of the permissions that SYSTEM has, it is often a target for attackers or penetration testers.

07. Introduction to Command Prompt (CMD)

Now that we have looked at ending startup processes, we can move on to learning about the Windows command line.

CMD and Environment (ENV) Variables

Explain to students that the Windows Command Prompt, also referred to as CMD or `cmd.exe`, is the command-line interface for Windows, comparable to a Unix shell, such as Bash for Linux.

Today's lesson will heavily use CMD. We'll get started by introducing some basic commands.

Have everyone launch CMD by going to the bottom-left Windows icon and typing "CMD". Right-click **Command Prompt** and select **Run as administrator**.

Explain that **environment variables** from the Bash programming module work the same way on Windows.

Environment variables, also called envvars, are special values that contain information about the system, such as the user's home directory, or the system's program files directory.

Envvars in Windows can be used for the following:

- Replacing long directory paths with shorter ones
- Grabbing the current time
- Finding the location of your system files

Environment variables differ from Linux's in that they are surrounded by percent signs (%) instead of starting with a \$ sign. For example:

- In Windows, the environment variable VAR would be: %VAR%.
- In Linux, the environment variable VAR would be: \$VAR.

Explain that we can combine environment variables with regular directory names.

- Run `cd %USERPROFILE%\Desktop` to navigate to the Desktop directory.

Explain that %USERPROFILE% is a variable assigned to the value of the current user's home directory, or in this case, C:\Users\azadmin. This is the same as \$HOME in Linux. Explain to students that, like in Linux, Windows has many default environment variables.

Mention a couple of the following default Windows environment variables:

Environment Variable	Default Value
%CD%	Current directory
%DATE%	Current date
%OS%	Windows
%ProgramFiles%	C:\Program Files
%ProgramFiles(x86)%	C:\Program Files (x86)
%TIME	Current time

%USERPROFILE%	C:\Users\{username}
%SYSTEMDRIVE%	C:\
%SYSTEMROOT%	C:\Windows

Reassure students that they won't need to know all of these and that a list of common envvars will be provided as reference whenever needed.

CMD Commands

Some commands are similar to Linux, while some are completely different. Cover the following navigation commands. We will practice them in the following demo.

- `cd` or `chdir` changes directories. This works like Linux's `cd`.
 - However, `cd` by itself will function like `pwd` (print working directory) in Linux.
- `dir` lists the contents of a directory. This works like Linux's `ls`.

Let's try these navigation commands with an environment variable.

- While still in `cd %USERPROFILE%\Desktop`, have everyone run the following:
 - `cd` to see the current directory
 - `cd ..` to move up to the parent directory
 - `dir` to list the contents of the directory, which should list the contents of `C:\Users\sysadmin`

Next, cover the following command prompt operations:

- `md` or `mkdir` creates directories.
- `copy` copies a file. This works like Linux's `cp`.
- `move` cuts and pastes files. This works like Linux's `mv`.
- `del` or `erase` deletes files and directories. Directories will prompt a user to confirm.

- Note that files deleted with these commands do not go to the Recycle Bin as they do when we delete files with a GUI. Deleted files cannot be restored.
- `rd` or `rmdir` removes a directory if it's empty. Non-empty directories must be removed with the `/s` parameter: `rmdir /s` or `rd /s`.
- `find` searches a file for specific criteria. For example, `find "hello" greeting.txt` will search the `greeting.txt` file for the string `hello`.
 - In this example, the first argument or parameter `"hello"` is the string to search for and the second argument or parameter is the path and file name.
 - Note that `find` is case sensitive while most command prompt commands are not case sensitive.
- `echo` prints output to the screen like in Linux.
- `type` followed by a file name shows the contents of a file. This is similar to `cat` in Linux.
- `cls` clears the screen of the CMD prompt.
- `exit` closes CMD, like in most terminals.

CMD Parameters and Help

Note that CMD uses what are known as "parameters," which can be considered the equivalent of "arguments" from our Bash lessons. The terminology "options" or "switches" may also be used to specify parameters with specific functionality.

The terminology you use isn't as important as understanding the function of a parameter. To find out more about a command, you can use `help <command>`.

- With CMD open, run `help del` and note the entries.

Deletes one or more files.

`DEL [/P] [/F] [/S] [/Q] [/A[:attributes]] names`

`ERASE [/P] [/F] [/S] [/Q] [/A[:attributes]] names`

names Specifies a list of one or more files or directories.

Wildcards may be used to delete multiple files. If a directory is specified, all files within the directory will be deleted.

/P Prompts for confirmation before deleting each file

/F Forces deleting of read-only files

/S Deletes specified files from all subdirectories

/Q Quiet mode, will not prompt for confirmation to delete on global wildcard

/A Selects files to delete based on attributes

R Read-only files S System files

H Hidden files A Files ready for archiving

I Not content-indexed files L Reparse points

O Offline files - Prefix meaning "not"

If Command Extensions are enabled, DEL and ERASE change as follows:

The display semantics of the /S switch are reversed in that it shows you only the files that are deleted, not the ones it could not find.

We can see here that the /f parameter "force" will cause del to delete a read-only file when used.

Explain that documentation can also be found by entering "cmd " into your search engine.

- For example, the first result for "cmd del" will likely be Microsoft's official online documentation for del.

Use CMD to Create To-Do List Demo

Let's try a few of these commands by using CMD to create a to-do list. We'll do this by outputting and appending printed console output to a text file. To start, have everyone run the following:

- `cd Desktop` to return to the `C:\Users\sysadmin\Desktop` directory.
- `mkdir Audit` to create an `Audit` subdirectory.
- `cd Audit` or `chdir Audit` to enter that subdirectory.

Explain that we can use `echo` with `>` in Windows to output the console text to a file. Have everyone run:

- `echo hello > todo.txt` to create a new file called `todo.txt`.
- `dir` to display the directory's contents, to confirm that the file exists.
- `type todo.txt` to show the contents of `todo.txt`, which is `hello`.
 - Reiterate that `type` works like `cat` in Linux.

Explain that if we use `echo` with `>` again, it will overwrite the file's contents. Have everyone run:

- `echo Review Task > todo.txt` to overwrite the contents of `todo.txt`.
- `type todo.txt` again to show that the contents of `todo.txt` are now `Review Task`.
-

Explain that we can use `>>` instead of `>` to append contents to a file without overwriting anything. Have everyone run:

- `echo 1. Check for CPU-Intensive Processes >> todo.txt` to append a to-do task to our `todo.txt` list.
- `echo 2. Disable Unnecessary Startup Processes >> todo.txt` to append a second to-do task to our to-do list.

- `type todo.txt` to show the new contents of the `todo.txt` file:

1. Check for CPU-Intensive Processes

2. Disable Unnecessary Startup Processes

Explain that we can use `del` to delete files. Have everyone run:

- `del todo.txt` to delete the `todo.txt` file.
- Confirm by running `dir` to show `todo.txt` has been deleted.
- Lastly, run `cls` to clear the terminal screen.

Explain that as students get more proficient with the command line, they will be more prepared to execute and automate tasks in future systems and security-related roles.

08. Intro to Task Manager and CMD Activity

- [Activity File: Task Manager and CMD](#)

09. Activity Review: Intro to Task Manager and CMD Activity

10. Break

11. Users, Groups, and Password Policies

Now we will use the command line to find user password policy and login information.

Introducing net

We will be using the command-line utility known as `net`.

- `net` allows an administrator to query and set various settings relating to user accounts, groups, and password policies.
 - For example, a system administrator might pull information about an employee who has left the company to see when they last logged in. They can then use `net` to delete their account.

We will be using the following `net` utilities:

- `net user`: For adding, removing, and managing users. In the following activity, we will use it to enumerate a list of all the users on the system, and then enumerate information about the `sysadmin` account.
- `net localgroup`: For adding, removing, and managing local groups.
 - While we are working with local groups now, we will be looking at domain groups in the future. Domain groups are groups created in Windows Server to manage entire sets of users with identical user permissions and policies.
- `net accounts`: Allows you to see password and login requirements for users to enforce password security policy.
 - `net accounts` can be used to set the following password policies:
 - The minimum number of characters required for a password.
 - The amount of time before passwords expire.
 - The minimum number of days before a password can be changed.
 - Number of times a password must be unique before it can be reused again. In other words, if this is set to two, a user needs to change their password to two new ones before an old one can be used again.

Managing these passwords is similar to editing `/etc/pam.d/common-password` settings in Ubuntu Linux.

Net Demonstration Setup

Next, we'll demonstrate `net` using the following scenario:

- Your CIO was curious about the groups and password policies on the Windows workstation. You are to retrieve more information from this Windows workstation using the `net user` command-line utility.

We will be using the `net` tool to find information about user groups and password policies by doing the following tasks:

- Enumerating users to see `net` output.
- Enumerating `azadmin`'s groups and password policies.
- Enumerating local groups with `net localgroup`.
- Enumerating the Windows workstation's current password policies with `net accounts`.

Enumerate Users with net

First, we will use the `net user` account to enumerate all the users on the system.

- Run `net user` and note the output:

```
Unset
User accounts for \\ML-RefVm-447521

-----
-----

Alex                azadmin             DefaultAccount
Guest               WDAGUtilityAccount

The command completed successfully.
```

Find azadmin's Password Status

Next, we will look at specific password settings for the `azadmin` user.

- Run `net user azadmin`. Note the following in the output:
 - **Password last set:** When the password was last set. For the user `azadmin`, it was last set on `3/20/2020 10:10:41 PM`.
 - **Note:** Your output will be different.
 - **Password expires:** The date that the current password for the user will expire. For `azadmin`, there is no expiration.
 - **Password changeable:** The earliest date that the user can change their password again. For `azadmin`, the date is `3/20/2020 10:10:41 PM`.
 - **Note:** Your output will be different.
 - **Password required:** The password policy that specifies whether the user needs a password. For this user, they *do* need a password.
 - **Note:** This is a given because `azadmin` is the primary local administrator account on this machine.

```
Unset
User name                azadmin

Full Name

Comment                  Built-in account for administering
the computer/domain

User's comment

Country/region code      000 (System Default)

Account active            Yes

Account expires           Never

Password last set        3/2/2020 10:10:41 PM

Password expires          Never
```

```
Password changeable          3/2/2020 10:10:41 PM
```

```
Password required            Yes
```

```
User may change password     Yes
```

```
Workstations allowed         All
```

```
Logon script
```

```
User profile
```

```
Home directory
```

```
Last logon                   8/24/2020 3:55:08 PM
```

```
Logon hours allowed          All
```

```
Local Group Memberships     *Administrators
```

```
Global Group memberships    *None
```

```
The command completed successfully.
```

- **Note:** Your output will look slightly different based on when you started.

Find Groups on the Machine

Now we will enumerate a list of groups on the Windows workstation. It's important to know how to enumerate groups on a system since groups are created, removed, and updated as often as users are.

Groups of users can include administrators, non-privileged users, users sorted by departments, users from third-party contractors, etc.

- Run `net localgroup` and note the output (your hostname will be different):

Unset

Aliases for \\ML-RefVm-447521

*Access Control Assistance Operators

*Administrators

*Backup Operators

*Cryptographic Operators

*Device Owners

*Distributed COM Users

*Event Log Readers

*Guests

*Hyper-V Administrators

*IIS_IUSRS

*Network Configuration Operators

*Performance Log Users

*Performance Monitor Users

*Power Users

*Remote Desktop Users

*Remote Management Users

```
*Replicator
*System Managed Accounts Group
*Users
The command completed successfully.
```

- **Note:** Your output will look slightly different.

Current Password Policy

Next, we'll pull the password policies that are currently set for the Windows workstation:

- Run `net accounts` and note the output:

```
Unset
Force user logoff how long after time expires?:      Never
Minimum password age (days):                        0
Maximum password age (days):                       42
Minimum password length:                            0
Length of password history maintained:               None
Lockout threshold:                                  Never
Lockout duration (minutes):                          30
Lockout observation window (minutes):                30
Computer role:                                       WORKSTATION
The command completed successfully.
```

- **Note:** Your output will look slightly different.

Enumerating information about users, groups, and password policies is important. This information helps your organization understand the current security policy posture on Windows workstations.

Additionally, if you need more information about `net` usage, you can use the command `net help` or `net help <user/accounts/localgroup>` in CMD and/or check out the following URLs:

- [net user](#)
- [net localgroup](#))
- [net accounts](#)

It is important for system administrators to quickly and reliably pull this type of information from workstations to ensure they adhere to company policies.

12. Activity: Users, Groups, and Password Policies

- [M7.1 Activity File: Users, Groups, and Password Policies](#)

13. Activity Review: Users, Groups, and Password Policies

14. Creating Users and Setting Password Policy

In this section, we will cover another common system administration responsibility: setting user password policies.

This section builds on the utilities `net user` and `net localgroup`.

We covered the importance of password policies in the previous Linux modules. Now we'll demonstrate applying similar concepts in Windows.

- We will create a new administrator and regular user on this machine.
- We will then set a password policy to ensure these users' passwords follow company-wide password policies.
 - As we saw at the end of the previous activity, there wasn't a minimum password length set. Every organization should have a minimum password length policy. We'll look at how to do that with the `gpedit` tool.

We'll use the `net user` utility to create Windows user accounts for the next two new users: senior developer Andrea and sales representative Barbara.

We are going to learn how to do the following:

- Create a regular user, `Barbara`.
- Create an administrator user, `Andrea`.
- Change user groups with `net localgroup`.
- Set workstation password policies with `gpedit`.

Creating a Regular User

For this demo, please use the Windows RDP host machine.

In the last activity, we used the command `net user` to show a list of all users on the system and then appended the results to our report.

Now we will use the `net user` utility again, adding a username after `net user`, and forward-slash-add (`/add`), so that our new command looks like this: `net user [username] /add`.

- We can also specify a password for the user by typing a password after the username, with the following syntax: `net user [username] [password] /add`.

To combine those steps and add a user named `Barbara` with the password `Coldcalls123!`, we use: `net user Barbara Coldcalls123! /add`.

- Run `net user Barbara Coldcalls123! /add`. If successful, we'll get the following output:

Unset

```
The command completed successfully.
```

- Now a regular user, `Barbara`, exists.

Creating an Administrative User with Elevated Privileges

Now we'll create a new user account for the new Windows senior developer, Andrea, and add it to the `Administrators` group.

Start by running the same command for creating a new user. You will use the name Andrea instead of Barbara:

- Run `net user Andrea JavaMaster123! /add` and note the output:

Unset

```
The command completed successfully.
```

- Note that you may see a message about Windows versions prior to Windows 2000 that could not support passwords longer than 14 characters. Press `Y`.

Unset

```
The password entered is longer than 14 characters. Computers  
with Windows prior to Windows 2000 will not be able to use  
this account. Do you want to continue this operation? (Y/N)  
[Y]: y
```

After creating the user `Andrea`, we can use the `net localgroup` command-line utility to add `Andrea` to the `Administrators` local group. This will allow `Andrea` to manage the Windows workstation as a super user.

- Run `net localgroup Administrators Andrea /add` and then note the output:

```
Unset
The command completed successfully.
```

We can use the `net user` command, as in the previous activity, to check the user's account parameters:

- Run `net user Andrea`. Note the `*Administrators` group at bottom of the output:

```
Unset
...[results truncated]

Local Group Memberships      *Administrators      *Users
Global Group memberships    *None

The command completed successfully.
```

We've now successfully created a new user and added them to the `Administrators` local group.

Setting the Windows Workstation's Password Policy

Now that we have our users, we want to define and enforce a strong password policy for this Windows computer.

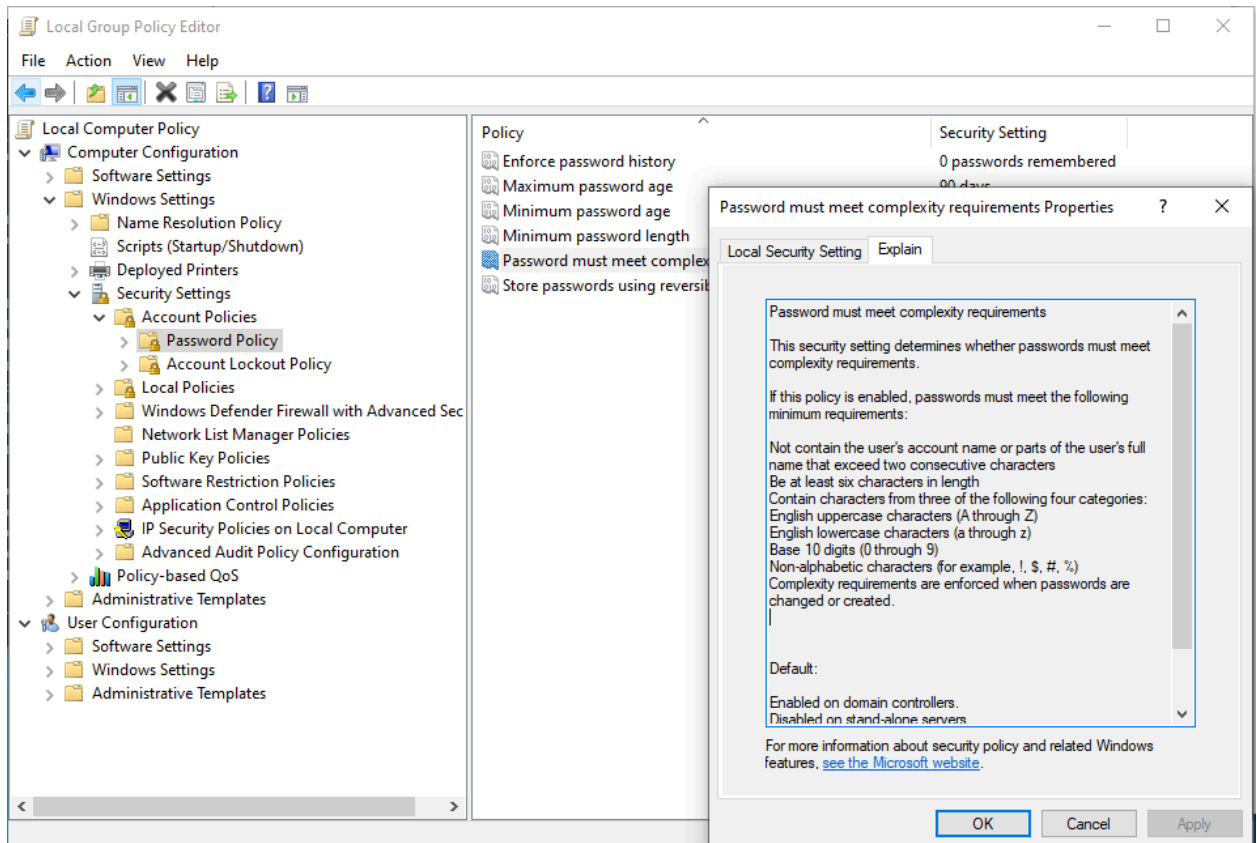
We will use the GUI tool `gpedit` to set password policies with the following parameters:

- Maximum password age to 60 days.
- Minimum password length to 8 characters.
- Password complexity requirements to be enabled.

To open **gpedit**, type "gpedit" in the Windows search bar and press Enter.

Under **Computer, Configuration**, navigate to **Windows Settings > Security Settings > Account Policies > Password Policy** and set the following:

- Set maximum password age to 60 days.
 - While **Maximum password age** is highlighted, right-click it and click **Properties** to see the tabs to enable and disable the option. You can also go to the **Explain** tab to see best practices for password expiration policies.



- Set the minimum password length to eight characters.
- Lastly, double-click **Password must meet complexity requirements**, and set to **Enabled**. Select **Apply**, then click the **Explain** tab and read the following:

Unset

Password must meet complexity requirements

This security setting determines whether passwords must meet complexity requirements.

If this policy is enabled, passwords must meet the following minimum requirements:

Not contain the user's account name or parts of the user's full name that exceed two consecutive characters

Be at least eight characters in length

Contain characters from three of the following four categories:

English uppercase characters (A through Z)

English lowercase characters (a through z)

Base 10 digits (0 through 9)

Non-alphabetic characters (e.g., !, \$, #, %)

Complexity requirements are enforced when passwords are changed or created.

Follow these same steps to create users and set password policies on the Windows workstation.

15. Activity: Create Users and Set a Password Policy

- [M7.1 Activity File: Creating Users and Setting Password Policies](#)

16. Activity Review: Create Users and Set a Password Policy

17. Task Scheduling

Task Scheduler is a GUI tool that allows system administrators to automate the execution of scripts and applications on a Windows system.

- These are similar to cron jobs in Linux. They can be set to execute at specific times or a certain amount of time after a user logs in.
- It is important to properly manage systems with scheduled tasks to automate security and system administration actions such as checking for updates for endpoint security software, sending important logs to systems such as SIEM (Security Information and Event Management), and scheduling system maintenance scripts.

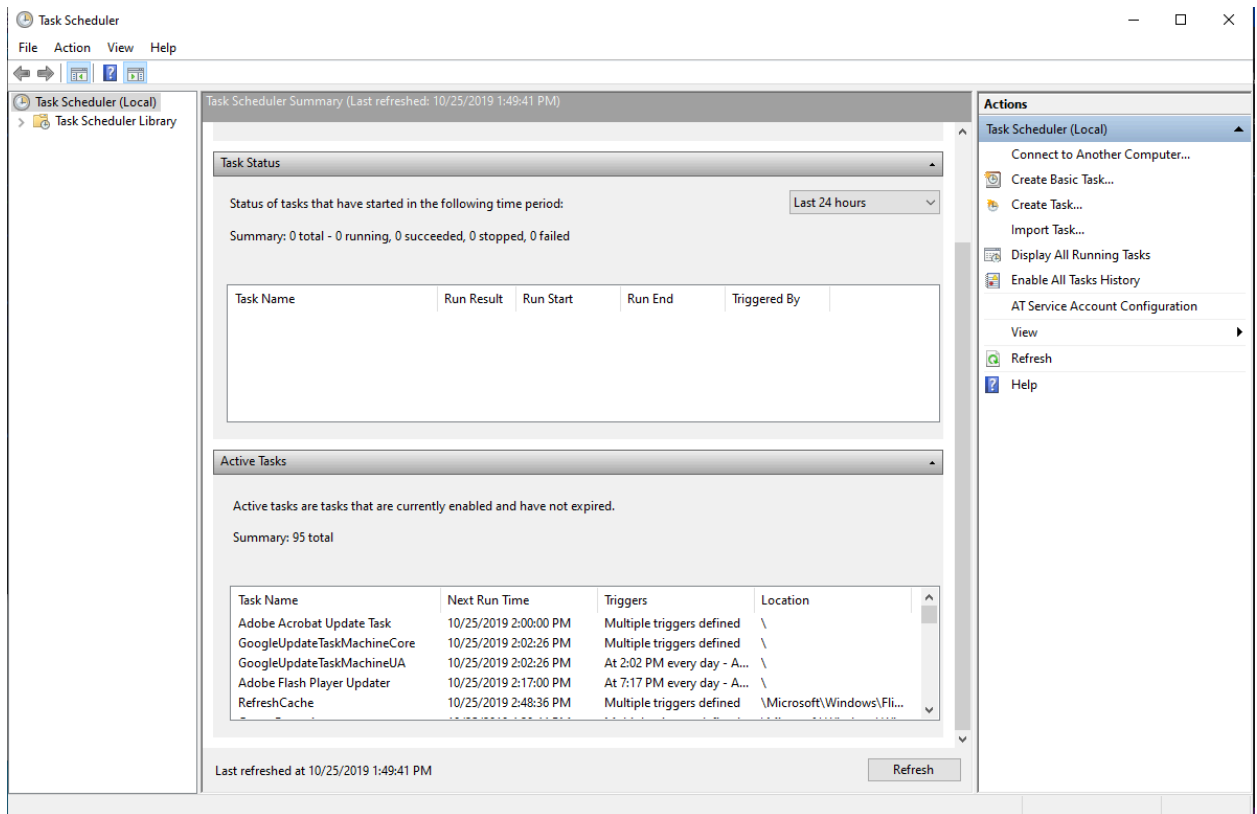
Task Scheduling Demo

Continue using the Windows RDP host machine.

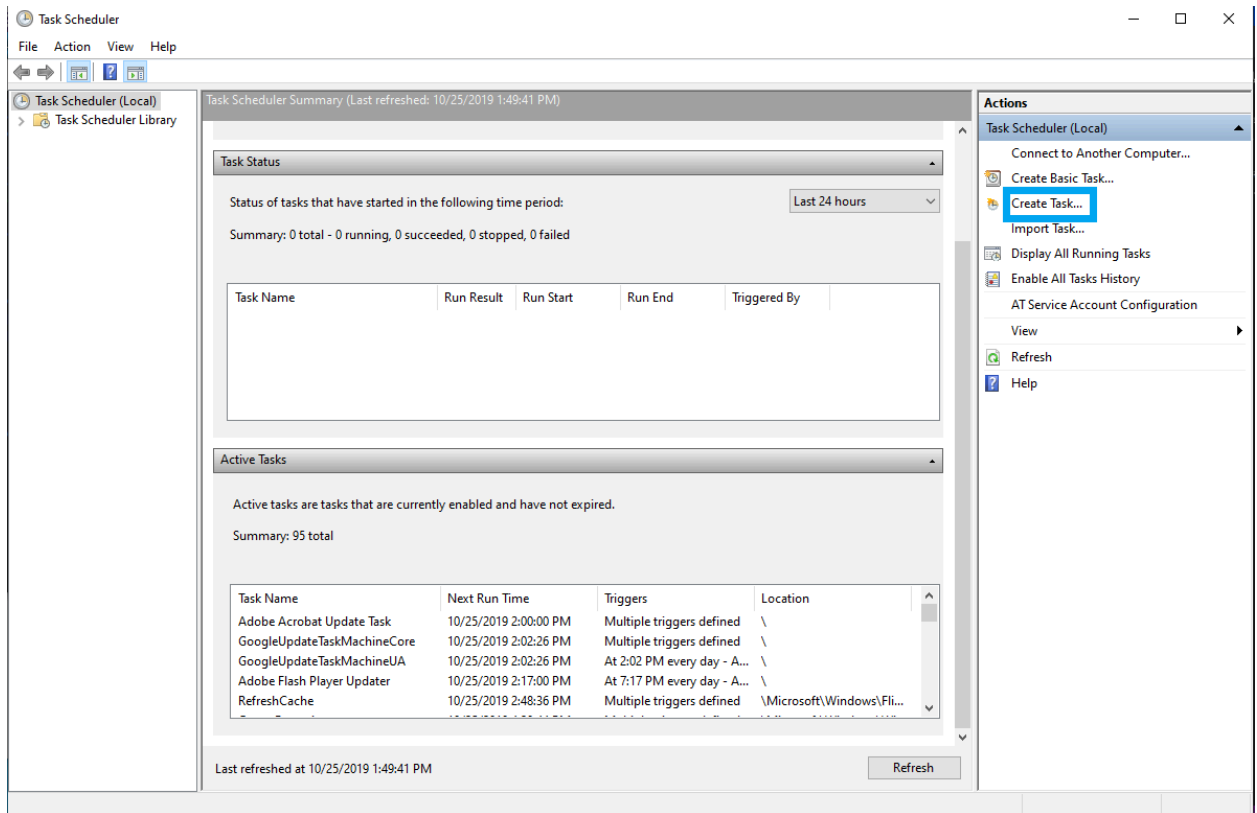
- Your CIO wants to schedule reports to be created on a daily basis. This system isn't remotely managed at the moment, and with a new developer coming in, huge changes might be made to the machine's configuration.
- This will help reduce workstation development drift for any new users for the Windows environment, and for this specific developer position and project.
- Your CIO has asked you to use the GUI tool Task Scheduler to create a task that will run each day.
-  M7.1 Activity File: Task Scheduling

Use the user `azadmin` to create a newly scheduled task.

- Click on the Windows search in the taskbar and search for "Task".
- Choose **Task Scheduler** from the list.



- Create a new task by selecting **Create Task...** on the **Actions** pane.
 - **Note:** If you don't see the Actions pane on the right, go to **View > Customize** and make sure Action Pane is enabled.



Setting the Task Trigger

- In the Create Task GUI window, navigate to the **Triggers** tab, and click **New....**
- Set **Begin the task** to **On a schedule**.
- Make sure your task is set to repeat daily with **Recur every:[x] day** set to **1**, and the start date set to today. (It should already be set, but you may have to input it manually).
- Finally, set the time to be a minute or two after the current time.
- Select **OK** to save your changes and return to the Create Task window.

Note: Remember to shut down your Hyper-V virtual machines and Windows RDP host machine.